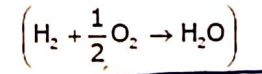


❖ Example Calculation of Thermochemical Equilibrium in a Rocket Chamber:

The rocket combustion chamber operates at $p = 210 \text{ atm}$ and an oxidizer-to-fuel mass ratio $\text{O/F} = 6$.

fuel rich

The stoichiometric ratio for full combustion



$\text{O/F} = 8$



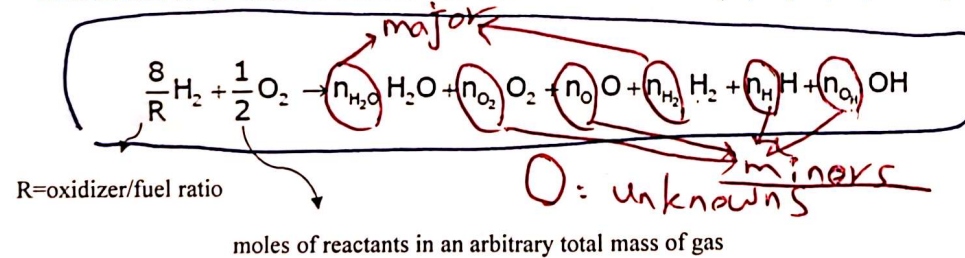
slightly fuel-rich and very high combustion temperature

I. Atomic Species Concentration

$$\text{st } R = r_{\text{st}} = \text{O/F} = \frac{\text{mass of O}_2}{\text{mass of fuel}} = \frac{\frac{1}{2} \times 32}{2 \times 1} = 8$$

Combustion of hydrogen in oxygen, with unknown amounts of water.

✓ The number of atomic moles for each kind of atom (O, H, N, C,) is the same before and after reaction



real reaction

$$\frac{8}{R}\text{H}_2 \text{ if } R = R_{\text{st}} \rightarrow \frac{8}{R} = 1$$

Here $R = 6$

$$N_{\text{O}_2} = 1/2$$

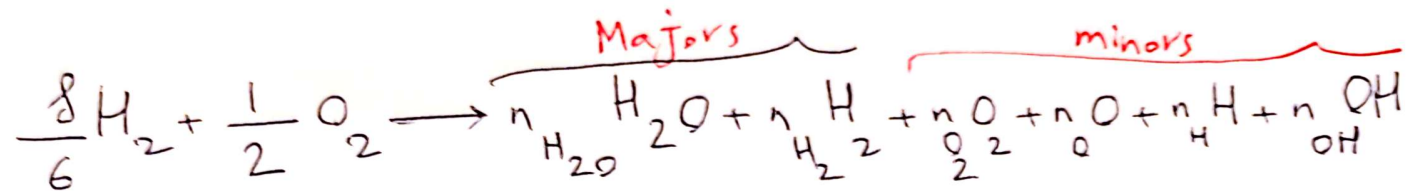
arbitrarily select

$$\frac{32 \times 1/2}{2N_{\text{H}_2}} = R$$

;

$$N_{\text{H}_2} = \frac{8}{R}$$

- The list of candidate reaction products is arrived at from our background knowledge of what species actually do occur.



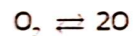
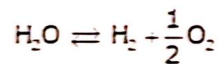
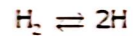
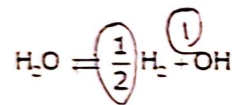
if R were low enough $\Rightarrow$ fuel-rich $\Rightarrow$ not include O₂ or O and oxygen species depleted.

- We would simply ignore them for practical purposes, as we ignore other exotic, but potentially real species, like, HO₂, H₃, etc.

$$\begin{aligned} \text{H-conservation: } & \frac{16}{R} = 2n_{H_2O} + 2n_{H_2} + n_H + n_{OH} \\ \text{O-conservation: } & 1 = n_{H_2O} + 2n_{O_2} + n_O + n_{OH} \end{aligned}$$

Two equations involving the six unknown

The equilibrium conditions (laws of mass-action) for four elementary reactions selected so as to involve these decomposition products, as well as some of the "major" products.



equilibrium relationship $\Rightarrow$

$$\frac{P_{H_2}^{1/2} P_{OH}}{P_{H_2O}} = K_1(T)$$

$$\frac{P_H^2}{P_{H_2}} = K_2(T)$$

$$\frac{P_{H_2} P_{O_2}^{1/2}}{P_{H_2O}} = K_3(T)$$

$$\frac{P_O^2}{P_{O_2}} = K_4(T)$$

$$P_{H_2O} = P \frac{n_{H_2O}}{n}$$

$$P_{O_2} = P \frac{n_{O_2}}{n}$$

$$P_{OH} = P \frac{n_{OH}}{n}$$

$$\vdots$$

$$\begin{aligned} (n = n_{H_2O} + n_{O_2} + n_{H_2} + n_H + n_{OH}) \\ \frac{P_{H_2O}}{P} = \frac{n_{H_2O}}{n} = \chi \\ \frac{P_{H_2}^{1/2} P_{OH}}{P_{H_2O}} = \frac{(P \frac{n_{H_2}}{n})^{1/2} (P \frac{n_{OH}}{n})}{P \frac{n_{H_2O}}{n}} = \frac{P^{1/2} n_{H_2}^{1/2} n_{OH}}{P n_{H_2O}} = \sqrt{\frac{n}{P}} K_1(T) \end{aligned}$$

rewrite (*) in terms of moles: $P_H^{1/2} P_{OH} = \frac{(P \frac{n_H}{n})^{1/2} (P \frac{n_{OH}}{n})}{P \frac{n_{H_2O}}{n}} = K_1(T)$

$$\Rightarrow \frac{P^{1/2} n_H^{1/2} n_{OH}}{n^{1/2} n_{H_2O}} = K_1(T) \Rightarrow \frac{P_{H_2O}}{n_H^{1/2} n_{OH}} = \sqrt{\frac{n}{P}} K_1(T)$$

$$K_1 = \exp \left(- \frac{\frac{1}{2} \mu_{H_2}^0 + \mu_{H_2O}^0 - \mu_{H_2O}^0}{RT} \right)$$

$$K_3 = \exp \left(- \frac{\mu_{H_2}^0 + \frac{1}{2} \mu_{O_2}^0 - \mu_{H_2O}^0}{RT} \right)$$

R=universal gas constant

μ = chemical potential of species i at 1 atm. pressure, at the given temperature T

$$K_2 = \exp \left(- \frac{2\mu_{H_2}^0 - \mu_{H_2}^0}{RT} \right)$$

$$K_4 = \exp \left(- \frac{2\mu_{O_2}^0 - \mu_{H_2}^0}{RT} \right)$$

We read $k_i(T)$ from tables

- The equilibrium constants still depend on temperature.

❖ Conservation of Energy

We need to know T $\sum_{\text{reactants}} n_i \bar{h}_i = \sum_{\text{products}} n_i \bar{h}_i$

Since the reactions occur at constant pressure, if we assume a well-insulated combustor, the enthalpy of the products must equal that of the reactants.

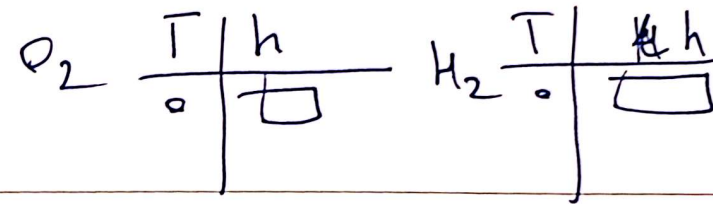
$$H_R = H_P$$

$$\frac{8}{R} h_{H_2}(T_i) + \frac{1}{2} h_{O_2}(T_i) = n_{H_2O} h_{H_2O}(T) + n_{O_2} h_{O_2}(T) + n_{H_2} h_{H_2}(T) + n_{H} h_H(T) + n_{OH} h_{OH}(T)$$

molar enthalpy of the species at temperature T

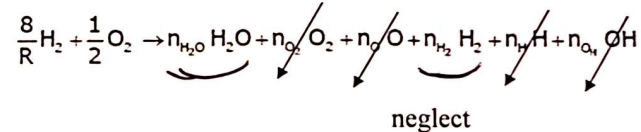
injection temperature

final equilibration temperature



❖ Numerical Iteration Procedure

$R = O/F = 6$, and $P = 210$ atm.



H-conservation:

$$\frac{16}{R}$$

$$= 2n_{\text{H}_2\text{O}} + 2n_{\text{H}_2} + n_{\text{H}} + n_{\text{OH}}$$

O-conservation:

$$1 = n_{\text{H}_2\text{O}} + 2n_{\text{O}_2} + n_{\text{O}} + n_{\text{OH}}$$

minors

$$2n_{\text{H}_2\text{O}} + 2n_{\text{H}_2} = \frac{8}{3}$$

$$n_{\text{H}_2\text{O}} = 1$$

$$n_{\text{H}_2\text{O}} = 1$$

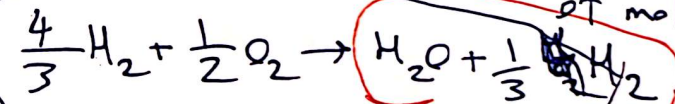
$$n_{\text{H}_2} = 1/3$$

$$n = 4/3$$

total number of moles

The enthalpy before reaction (assuming very cold reactants $T_i = 0^\circ \text{K}$):

$$H = \frac{4}{3} h_{\text{H}_2}(\text{O}) + \frac{1}{2} h_{\text{O}_2}(\text{O}) = -15,632 \text{ J (for the } \frac{4}{3} \text{ moles of H}_2, \frac{1}{2} \text{ mole of O}_2 \text{).}$$



$$H_R = H_P$$

$$H_P = -15,632 \text{ J}$$

For the products to have the same enthalpy, the temperature must be very high. Using the table in Ref. 12.1 (pp. 692-693):

T(K)	3400	4000	4200
$h_{\text{H}_2\text{O}}$ (J/mole)	-92,973	-58,547	-46,924
h_{H_2} (J/mole)	103,738	126,846	134,700
$h_{\text{H}_2\text{O}} + \frac{1}{3} h_{\text{H}_2}$	-58,394	-16,265	-2,024

Linear interpolation between 4000 and 4200 K

$$-15,632 \text{ J}$$

$$T = 4009 \text{ K}$$

<https://webbook.nist.gov/cgi/cbook.cgi?ID=C1333740&Mask=1&Type=JANAFG&Table=on>

$$T = 4009 \text{ K} \rightarrow \text{read}$$

$$\begin{cases} k_1(T) \\ k_2(T) \\ k_3(T) \end{cases} \quad k_4(T)$$

in Table 12 of Ref. 12.1. With some interpolation

$$n_{OH} = \frac{n_{H_2O}}{n_{H_2}^{1/2}} \sqrt{\frac{n}{p}} K_1 = \frac{1}{\sqrt{1/3}} \sqrt{\frac{4/3}{200}} 0.974 = 0.138$$

$$K_1 = 0.974 \text{ (atm)}^{1/2}$$

$$K_2 = 2.61 \text{ (atm)}$$

$$n_H = \sqrt{\frac{nn_{H_2}}{p}} K_2 = \sqrt{\frac{4/3 \times 1/3}{200}} \times 2.61 = 0.076$$

$$K_3 = 0.589 \text{ (atm)}^{1/2}$$

$$K_4 = 2.286 \text{ (atm)}$$

$$n_{O_2} = \left(\frac{n_{H_2O}}{n_{H_2}} \right)^2 \frac{n}{p} K_3^2 = \left(\frac{1}{1/3} \right)^2 \frac{4/3}{200} (0.589)^2 = 0.021$$

$$n_O = \sqrt{\frac{nn_{O_2}}{p}} K_4 = \sqrt{\frac{4/3 \times 0.021}{200}} \times 2.286 = 0.018$$

End of the first loop

❖ Atom conservation equations including the minor species:

$$n_{H_2O} + n_{H_2} = \frac{4}{3} - \frac{0.076 + 0.138}{2} = 1.226$$

H-Conservation

$$n_{H_2O} = 1 - 2 \times 0.021 - 0.018 - 0.138 = 0.802$$

O-Conservation

$$n_{H_2} = 1.226 - 0.802 = 0.424$$

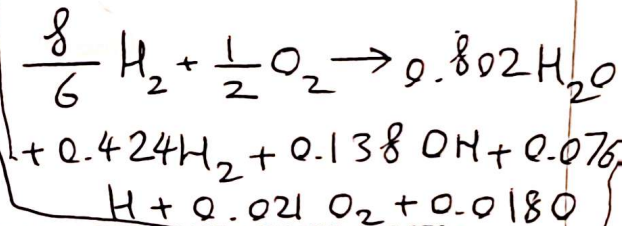
$$n = 1.226 + 0.138 + 0.076 + 0.021 + 0.018 = 1.480$$

$$h(2800 \text{ K}) = 0.802 \times (-126,533) + 0.424 \times 81,370 + 0.138 \times 121,729 + 0.076 \times 269,993 + 0.021 \times 90,144 + 0.018 \times 301,587 = -22,338 \text{ J}$$

$$h(3000 \text{ K}) = 0.802 \times (-115,466) + 0.424 \times 88,743 + 0.138 \times 129,047 + 0.076 \times 274,148 + 0.021 \times 98,098 + 0.018 \times 305,771 = -8,769 \text{ J}$$

Since we still want $h = -15,632 \text{ J}$, linear interpolation gives a temperature

$$T = 2800 + \frac{200}{-8,769 + 22,338} (-15,632 + 22,338) = 2899 \text{ K}$$



✓ The partial decomposition of H_2O and H_2 into the minor species is an endothermic process, the new temperature will be lower.

H-conservation

$$\frac{8}{3} = 2n_{H_2O} + 2n_{H_2} + n_H + n_{OH}$$

$$2(n_{H_2O} + n_{H_2}) = \frac{8}{3} - n_H - n_{OH}$$

$$n_{H_2O} + n_{H_2} = \frac{1}{2} \left(\frac{8}{3} - 0.076 - 0.138 \right) = 1.226$$

O-conservation

$$1 - n_{H_2O} + 2n_{O_2} + n_O + n_{OH} \Rightarrow n_{H_2O} = 1 - 2n_{O_2} - n_O - n_{OH}$$

$$1 - 2(0.021) - 0.018 - 0.138$$

$$n_{H_2O} = 0.802$$

$$\Rightarrow n_{H_2O} + n_{H_2} = 1.226 \Rightarrow n_{H_2} = 1.226 - 0.802 =$$

$$0.424$$

4 minors known

Solve 2 balance Eq^s

to get majors

Subject:

Year:

Month:

$$y_{A_{20}} = y_1$$

$$s = y_m$$

$$r_{O_2} = y_2$$

$$-22222 x_1$$

$$-10222 x_m$$

$$-15229 x_2$$

$$y_m y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x_m - x_1)$$

$$y_m = \frac{r_{A_{20}} + r_{O_2}}{-15229 + 22222} (-10222 + 22222)$$

$$y_m = 2199 \text{ k}$$

"Gas Eq"

- Accelerate convergence by adopting as a new trial temperature the average of the last two :

$$T = \frac{\textcircled{1} 4009 + \textcircled{2} 2899}{2} = 3454 \text{ K}$$

- The new equilibrium constants (at 3480 K):

$$K_1 = 0.251 (\text{atm})^{1/2}$$

$$K_2 = 0.316 (\text{atm})$$

$$n_{\text{OH}} = 0.0250$$

$$n_{\text{H}} = 0.0297$$

$$n_{\text{O}_2} = 0.0079$$

$$n_{\text{O}} = 0.00105$$

$$K_3 = 0.184 (\text{atm})^{1/2}$$

$$K_4 = 0.454 (\text{atm})$$

ITERATION	$N_{\text{H}_2\text{O}}$ (mol/kg)	N_{O_2}	N_{O}	N_{H_2}	N_{H}	N_{OH}	T (K)	$\frac{T + T_{\text{OLD}}}{2}$	K_1 (atm) ^{1/2}	K_2 (atm)	K_3 (atm) ^{1/2}	K_4 (atm)
1	53.57	0	0	17.86	0	0	4009		0.974	2.610	0.589	2.286
2	42.96	1.125	0.964	22.71	4.071	7.393	2899	3454	0.236	0.281	0.173	0.190
3	52.09	0.423	0.056	17.87	1.591	1.339	3784	3619	0.356	0.587	0.259	0.432
4	50.42	0.199	0.172	18.76	1.912	2.587	3644	3632	0.370	0.619	0.267	0.458
5	50.48	0.182	0.171	18.66	2.025	2.555	3642	3637				